Haemoglobin

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Identity Sheet

Property	Value
Name	Haemoglobin
Other names	Hemoglobin, Hb, Hgb
Chemical formula	$C_{2952}H_{4664}N_{812}O_{832}S_8Fe_4$ (approximate for HbA tetramer)
Molecular weight	~64,458 Da (HbA tetramer: 2α + 2β globin chains)
PubChem CID	49870888 (heme group: CID 26945 for protoporphyrin IX-Fe)
Appearance	Red protein; oxyHb bright red, deoxyHb dark red-purple
Isoelectric point	~6.8–7.1
Prosthetic group	Heme (iron-protoporphyrin IX), one per globin subunit (4 total)

Structural formula (heme group):

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N——N
/ Fe  / |  N——+——N
(4 pyrrole rings
with Fe2+ centre)
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Haemoglobin is a **globular metalloprotein** (~64.5 kDa) composed of four globin subunits, each containing one heme prosthetic group. The heme iron (Fe^{2+}) reversibly binds oxygen. Haemoglobin is normally **intracellular** within RBCs. Its presence as a free molecule in urine (haemoglobinuria) indicates intravascular haemolysis or RBC lysis within the urinary tract. It exhibits characteristic optical properties: intense **Soret absorption band at ~415 nm** and Q bands at 540 and 577 nm (oxyHb).

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Myoglobin	$C_{769}H_{1212}N_{210}O_{218}S_5Fe$	17,800	Single heme group; monomeric; from muscle injury (rhabdomyolysis); MW much lower
Free heme	$C_{34}H_{32}FeN_4O_4$	616.49	Prosthetic group alone; highly toxic, pro-oxidant
Bilirubin	$C_{33}H_{36}N_4O_6$	584.66	Heme degradation product; linear tetrapyrrole; yellow

Molecule	Formula	MW (g/mol)	Key Difference
Intact RBCs	—	—	Cellular form; dipstick detects both; microscopy distinguishes intact cells from free Hb

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Medical Information

Origin

Endogenous: Biosynthesis — Erythropoiesis Haemoglobin is synthesised in **erythroid precursors** in the bone marrow during erythropoiesis. Globin chains (α, β) are synthesised on ribosomes. Heme is synthesised via the heme biosynthetic pathway (see Porphobilinogen sheet). Iron is inserted into protoporphyrin IX by ferrochelatase. Each RBC contains ~270 million Hb molecules. Total body Hb: ~750 g in adults. Daily Hb synthesis: ~6 g/day (replacing ~0.8% of circulating RBCs).

Exogenous No dietary source of intact haemoglobin contributes to circulating Hb. Dietary iron (heme iron from meat, non-heme from plants) is a critical raw material. Blood transfusion introduces donor Hb/RBCs. Recombinant Hb-based oxygen carriers (HBOCs) are experimental.

Primary & Secondary Biological Roles

Primary role: - **Oxygen transport:** Hb carries O_2 from lungs to tissues (cooperative binding, sigmoid O_2 dissociation curve, P50 ~26 mmHg).

Secondary roles: - **CO_2 transport:** ~15–20% of CO_2 carried as carbaminohaemoglobin. - **pH buffering:** Histidine residues buffer H^+; Bohr effect links pH to O_2 release. - **Nitric oxide (NO) processing:** DeoxyHb reduces nitrite to NO (vasodilator); oxyHb scavenges NO.

Catabolism and Elimination Pathway

- Aged RBCs are phagocytosed by splenic/hepatic macrophages.
- Hb → globin (recycled to amino acids) + heme → biliverdin → **bilirubin** (conjugated in liver, excreted in bile).
- Iron is recycled via ferritin/transferrin.
- Free plasma Hb (from intravascular haemolysis) is bound by **haptoglobin** → cleared by hepatic macrophages.
- When haptoglobin is saturated, free Hb is **filtered by the glomerulus** (Hb dimers ~32 kDa pass the glomerular barrier). Tubular cells reabsorb Hb (megalin/cubilin receptors) and catabolise it.
- When reabsorptive capacity is exceeded → **haemoglobinuria** (free Hb in urine).
- Minor route: Hb → methaemalbumin (bound to albumin in plasma).

Expression in Humans

Normal Levels

Compartment	Reference Range
Blood Hb	Male: 13.5–17.5 g/dL; Female: 12.0–16.0 g/dL
Plasma free Hb	< 5 mg/dL (normally negligible)
Urine Hb (free)	Absent (negative on dipstick)
Urine RBC/Hb (dipstick)	Negative

Factors Influencing Levels **Increased Hb in urine (haemoglobinuria):** - Intravascular haemolysis (autoimmune haemolytic anaemia, PNH, TTP, HUS, malaria) - Transfusion reactions (ABO incompatibility) - Mechanical haemolysis (prosthetic heart valves, march haemoglobinuria) - Severe burns - G6PD deficiency (drug/food-triggered haemolysis) - Paroxysmal cold haemoglobinuria - In-vitro RBC lysis in urine (alkaline/hypotonic urine)

Absent (normal): - Intact RBC membranes, functional haptoglobin system - No haemolytic process - Adequate haptoglobin capacity

Associated Pathologies

Condition	Urine Hb	Key Symptoms
Paroxysmal nocturnal haemoglobinuria (PNH)	Dark morning urine, free Hb	Haemolysis, thrombosis, pancytopenia; rare (~1–10/million
Autoimmune haemolytic anaemia	Haemoglobinuria if severe	Anaemia, jaundice, fatigue
Transfusion reaction	Acute haemoglobinuria	Fever, flank pain, dark urine, DIC
Rhabdomyolysis (myoglobin)	Dipstick positive but myoglobin not Hb	Muscle pain, dark urine, AKI risk
March haemoglobinuria	Transient, exercise-related	Benign; after prolonged marching/running

Presence in Urine

Should it be normally present? No — urine should be free of haemoglobin. Any detection indicates intravascular haemolysis (free Hb) or RBC lysis within the urinary tract.

Normal urinary levels: Undetectable (negative dipstick, <0.03 mg/dL).

Form in urine: Free haemoglobin is present as **Hb dimers** (~32 kDa, αβ pairs) or tetramers; also oxidised forms: **methaemoglobin** (Fe^{3+}). In aged samples, Hb degrades to heme and globin.

Pathological significance:

Urinary Hb	Possible Causes	Prevalence
Present (haemoglobinuria)	Intravascular haemolysis, severe haematuria with RBC lysis	Variable; PNH ~1:100,000
Absent	Normal	—

Solubility: Haemoglobin is soluble in water/urine at physiological concentrations. At very high concentrations, Hb can precipitate in renal tubules (Hb cast nephropathy), especially at acidic pH.

Detection in Urine

Available Clinical Assays

1. **Urine dipstick (pseudoperoxidase reaction):**
   - **Principle:** Hb (and myoglobin) catalyses oxidation of TMB (3,3',5,5'-tetramethylbenzidine) by organic peroxide → green-blue colour.
   - **Detection:** Colour change graded trace to 3+. Intact RBCs produce spotted pattern; free Hb produces uniform colour.
   - **LOD:** ~0.015–0.062 mg/dL free Hb (~5–10 intact RBC/μL).

- **Absorption:** TMB oxidised product absorbs at ~660 nm.
- **Advantages:** Rapid, inexpensive, very sensitive.
- **Disadvantages:** Does not distinguish Hb from myoglobin. False positives from oxidants.

2. **Spectrophotometry (Soret band):**
   - **Principle:** Direct UV-Vis measurement of free Hb in centrifuged urine supernatant.
   - **Detection:** Absorbance at **415 nm** (Soret band).
   - **LOD:** ~0.5 mg/dL.
   - **Advantages:** Quantitative, specific for Hb (vs myoglobin at 409 nm).
   - **Disadvantages:** Requires spectrophotometer; not point-of-care.
3. **Immunoturbidimetry / immunonephelometry:**
   - **Principle:** Anti-Hb antibodies cause agglutination proportional to Hb concentration.
   - **Detection:** Turbidity at 340 nm.
   - **LOD:** ~0.01 mg/dL.
   - **Advantages:** Specific for Hb (not myoglobin).
   - **Disadvantages:** Costly, specialised.

Optimal Urine Type for Measurement

- **Fresh midstream urine** — RBCs lyse over time in dilute/alkaline urine, converting haematuria to haemoglobinuria.
- Analyse within **1–2 hours** of collection.
- **First morning urine** is more concentrated but may show more RBC lysis.
- Centrifuge to distinguish: **pink supernatant** = free Hb; **clear supernatant + red pellet** = intact RBCs.
- Protect from light (methaemoglobin formation).

Actual Gold Standard

The **urine dipstick** is the primary screening tool. For confirmation and quantification of free Hb, **UV-Vis spectrophotometry at 415 nm** (Soret band) on centrifuged supernatant is the reference method. The Harboe method (1959) measures Hb at 415 nm with correction for turbidity at 380 and 450 nm. Immunoassays provide the highest specificity (Hb vs myoglobin).

Interferences in Measurement

Interference	Effect	Mechanism
Myoglobin	False positive (dipstick)	Same pseudoperoxidase activity; MW 17.8 kDa also passes glomerulus
Ascorbic acid	False negative	Reduces oxidised TMB chromogen
Hypochlorite (bleach)	False positive	Chemical oxidation of TMB
Bacterial peroxidases	False positive (rare)	Mimic pseudoperoxidase activity
High-dose captopril	False negative	Thiol groups interfere
Formaldehyde (preservative)	False negative	Cross-links Hb, reduces peroxidase activity
High specific gravity	Decreased sensitivity	Concentrated urine inhibits chromogen development
Delay in analysis	Hb degradation	MetHb and heme formation reduce peroxidase activity

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis (Soret band):** OxyHb: **415 nm** ($\varepsilon \approx 131{,}000\ M^{-1}cm^{-1}$ per heme), 542 nm, 577 nm. DeoxyHb: 430 nm, 555 nm. MetHb: 405 nm, 500 nm, 630 nm.
  - LOD: ~0.1 mg/dL (~60 nM tetramer).
  - Sample prep: Centrifuge; measure supernatant; correct for turbidity.
- **NIR:** Weak absorption at ~760 nm (deoxyHb), ~940 nm (water interference). Not practical for urine.
  - LOD: >10 mg/dL.

Fluorescence Detection

- **Intrinsic porphyrin fluorescence:** Hb fluorescence is heavily **quenched by iron** — very weak (quantum yield <0.01%).
- **Zinc protoporphyrin (ZnPP):** If heme iron is replaced by zinc (as in iron deficiency), ZnPP fluoresces strongly (Ex 425 nm / Em 594 nm). Not relevant for urinary Hb.
- **Indirect methods:** Heme extraction → demetallation → free protoporphyrin fluorescence (Ex 405 nm / Em 630 nm). LOD: ~10 nM heme.
- **Fluorescent anti-Hb immunoassays:** FITC-labelled antibodies; Ex 490 / Em 520 nm. LOD: ~0.001 mg/dL.
- **Sample prep:** For direct methods, centrifugation; for immunoassays, standard ELISA/lateral flow protocols.

Raman Detection Haemoglobin has a rich resonance Raman spectrum:

Peak (cm^{-1})	Assignment
~675	Pyrrole deformation (ν_7)
~750	Pyrrole breathing
~1003	Phenylalanine
~1130	Methine C–H deformation
~1375	Oxidation state marker (ν_4) — Fe^{2+} vs Fe^{3+}
~1550–1580	Spin/ligation state marker (ν_2)
~1620	Vinyl C=C stretch
~1640	Oxygenation marker (ν_{10})

- **Resonance Raman** with excitation at **405 nm (Soret)** or **532 nm (Q-band)**: enhancement factor 10^3–10^6 for porphyrin modes.
- LOD: ~1 μM Hb in buffer; ~10 μM in urine (with background subtraction).
- Can distinguish oxyHb, deoxyHb, metHb by oxidation/spin markers.
- **Sample prep:** Centrifuge urine; measure supernatant or concentrate.

FTIR Detection

Band (cm^{-1})	Assignment	Notes
~1650	Amide I (α-helix of globin)	Dominant protein secondary structure band
~1540	Amide II	N–H bend + C–N stretch
~1450	CH_2/CH_3 deformation	Side chains

Band (cm^{-1})	Assignment	Notes
~1230–1240	Amide III	—
~1080	C–O stretch (carbohydrate, if glycated)	Relevant for HbA1c

- ATR-FTIR can detect Hb at >1 mg/dL concentrations in dried samples. Not sensitive enough for typical haemoglobinuria levels.
- **Sample prep:** Centrifuge supernatant, dry on ATR crystal.
- LOD: ~1 mg/dL (~0.6 μM tetramer).

Voltammetry Detection

- **Heme iron electrochemistry:** Direct electron transfer at modified electrodes. Fe^{3+}/Fe^{2+} reduction at **-0.2 to -0.4 V vs Ag/AgCl** (pH 7, carbon nanotube/graphene electrodes).
- **Amperometric detection:** Applied potential at -0.3 V; current proportional to Hb concentration.
- LOD: ~0.01–0.1 mg/dL on nanostructured electrodes.
- Linear range: 0.1–100 mg/dL.
- **Sample prep:** Centrifuged supernatant; buffer pH adjustment.

Other Detection Technologies

1. **Lateral flow immunoassay:** Anti-Hb antibody on nitrocellulose strip with gold nanoparticle label. LOD: ~0.01 mg/dL. Distinguishes Hb from myoglobin. 10 min. Point-of-care.
2. **Smartphone-based colorimetry:** Image analysis of dipstick colour using phone camera + app. LOD: comparable to visual dipstick (~0.015 mg/dL). Improved objectivity.
3. **Microfluidic Hb sensors:** Lab-on-chip with integrated Soret-band absorbance measurement (415 nm LED + photodiode). LOD: ~0.1 mg/dL. μL sample volumes.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Dipstick (TMB peroxidase)	~0.015 mg/dL	~660 nm product	None	Rapid, sensitive	Cross-reacts with myoglobin
UV-Vis spectrophotometry	~0.1 mg/dL	415 nm (Soret)	Centrifuge	Quantitative, Hb-specific	Instrument needed
Immunoturbidimetry	~0.01 mg/dL	340 nm (turbidity)	None	Hb-specific	Costly
Resonance Raman	~1 μM	Ex 405/532 nm; 1375 cm^{-1}	Centrifuge	Oxidation state info	Specialised
Fluorescence (immunoassay)	~0.001 mg/dL	Ex 490 / Em 520 nm	ELISA protocol	Ultra-sensitive, specific	Labour-intensive
FTIR	~1 mg/dL	1650 cm^{-1} (Amide I)	Dry sample	Multianalyte	Low sensitivity
Voltammetry (DET)	~0.01 mg/dL	-0.3 V vs Ag/AgCl	Buffer adjustment	Sensitive, low-cost	Selectivity challenges

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Lateral flow (anti-Hb)	~0.01 mg/dL	Gold NP colour	None	POC, Hb vs myoglobin	Semi-quantitative

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